CURBCOUNT · REGULATORY AND STANDARDS REGISTER

CurbCount regulatory and standards register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Opened in the kit 1.9.1 upgrade

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The design is a privacy-first counter, so the rules that matter most are data protection and radio use; the radar option adds a 60 GHz transmitter whose rules are still to assess. This register is not legal advice: anything marked "to assess" needs a qualified reviewer before any product claim. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	Regulation (EU) 2016/679 (GDPR), Article 25, data protection by design and by default	European Union	Privacy built into the design; processing only what is needed	No image, audio or identifier leaves the device and thermal frames are held in RAM only (PR-004, STM S1 and S20). Whether a count record is personal data at all is to assess	Design lead; data protection adviser	Met by design; legal view to assess
G2	Regulation (EU) 2024/1689 (AI Act)	European Union	Rules on AI systems and prohibited practices, including real-time remote biometric identification in public spaces	The counter identifies no one and cannot resolve a face (7.6 px/m). Classification of the tracker under the Act is to assess	Design lead; regulatory consultant	To assess
G3	IEC 62676-4 (video surveillance systems, application guidelines), pixel density levels	International	Pixels per metre needed to detect, recognise or identify a person	Used as the yardstick for PR-004: 7.6 px/m against 25 px/m to detect and 250 to identify. Not a certification	Design lead	Met by design (CAL-001 B)
G4	EU licence-free 863 to 870 MHz band and its 1 % duty-cycle rule for LoRaWAN uplinks	European Union	Airtime limit per device per hour	0.82 s/h at SF9 against 36 s/h (PR-018). The governing ETSI standard and the regional LoRaWAN parameters are researched once in the FieldNode repo; the number is not repeated here	Design lead; FieldNode project	Met by design (CAL-001 G); standard number to assess
G5	US licence-free 902 to 928 MHz band under 47 CFR Part 15 and the 400 ms dwell limit in the LoRaWAN US915 plan	United States	Dwell time and power of uplinks	371 ms at DR0 against 400 ms (PR-018). Section of Part 15 and any certification of the radio module to assess; FieldNode owns the radio	Design lead; FieldNode project	Met by design (CAL-001 G); certification needed later
G6	IEC 60529 (IP code), degrees of protection of enclosures	International	Ingress of dust and water	IP65 target for the enclosure and the gasketed head (PR-011). Tested at TRL 4, not now	Design lead	Applies; test at TRL 4

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G7	Lithium iron phosphate cells: cell safety and transport rules (cell maker datasheet; transport classification of lithium cells)	International	Charge limits, protection and shipping of cells	One 6 Ah cell with a fused holder and NTC lockout (BOM lines 2 and 3); charging only above 0 degrees C (STM S8). Standard numbers for the cell and for transport to assess	Design lead; FieldNode project	To assess
G8	Permission to attach equipment to a street pole	Local, set by the pole owner (city, utility or lighting authority)	Who may fit what to a pole, load and electrical rules	Clamp-on, no drilling or wiring (PR-014); the owner's attachment rules and any structural sign-off are to assess with the first partner	Partner city; asset owner	To assess
G9	60 GHz radar emission rules (option B only)	Each country of use	Transmit power, bandwidth and certification for short-range radar in the 60 GHz band	Option B adds a 60 GHz transmitter. The governing rules in the EU, the US and Canada are not researched or cited here; do not claim compliance	Design lead; regulatory consultant	To assess (option B only)
G10	ICNIRP 2020 guidelines for limiting exposure to electromagnetic fields, 100 kHz to 300 GHz	International	Public exposure limits (10 W/m2 above 6 GHz)	Option B: about 12 mW/m2 at head height on a 1 W overestimate (STM S21), far below the limit. Value to confirm against the guideline text	Design lead	Met by design on a first-order check (option B only); to confirm
G11	Local wind-load and roadside-object rules (height above the roadway, clearance from the carriageway)	Local	Loads on poles; clearances over road and sidewalk	Head at 4.3 m, 35 m/s gust check (PR-015, CAL-001 H). Local codes to assess with the partner	Partner city; asset owner	To assess
G12	Local rules on signage and notice of public sensing	Local	Whether and how a public notice is required	A notice plate on every pole (PR-019, CBC-DDR-001 D7). Whether the wording meets local rules is to assess	Partner city; legal adviser	To assess
G13	Substance and waste rules for electronics (for example the EU RoHS directive 2011/65/EU)	European Union	Restricted substances in electrical equipment	Applies to whoever places an assembled counter on the market; an open design published as files is	Builder	To assess if sold

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
				not that. Bought parts are the maker's choice		

Status: to assess, applies, does not apply, met by design, certification needed later.

Notes

- Standards shared across the FieldNode family (radio, cells, enclosure) are researched once in the FieldNode repo and are not copied here; this register names them and states how CurbCount meets them.
- No standard number, classification or limit in this register was invented: where the governing number was not confirmed in this session, the row says "to assess".
- Option B (G9, G10) applies only if the radar head is carried forward.

Sources

- Regulation (EU) 2016/679, https://eur-lex.europa.eu/eli/reg/2016/679/oj, as cited in CBC-PRB-001.
- Regulation (EU) 2024/1689, https://eur-lex.europa.eu/eli/reg/2024/1689/oj, as cited in CBC-PRB-001.
- ICNIRP, Guidelines for limiting exposure to electromagnetic fields (100 kHz to 300 GHz), *Health Physics* 118(5), 2020.
- FieldNode repository documents for radio and cell rules (not copied).
- The remaining rows cite the rule by name only; their texts were not read in this session, hence "to assess".